Legislative Role Models: Female Ministers, Participation, and Influence in Parliamentary Debate

Jack Blumenau

University College London

February 27, 2018

Motivation

Barbara Castle

Motivation

Barbara Castle

Betty Boothroyd

"Many years ago I worked in the House of Commons for a woman that I admired very much called Barbara Castle. She was my role model because I felt, well, if Barbara can do it then I can do it."

Betty Boothroyd, Speaker of the House of Commons.

Research question

When women are promoted to high political office, do they serve as role models to other women in politics?

Overview

- **Context:**
  - Female cabinet ministers in the House of Commons (1997 – 2017)
  - Behaviour of other female MPs in parliamentary debate
- **Data and strategy:**
  - Half a million speeches
  - Leverage within-ministry changes in minister gender over time
- **Results:**
  - Female ministers increase the **participation** of other female MPs
  - Female ministers increase the **influence** of other female MPs

Outline

Motivation

Data and empirical strategy

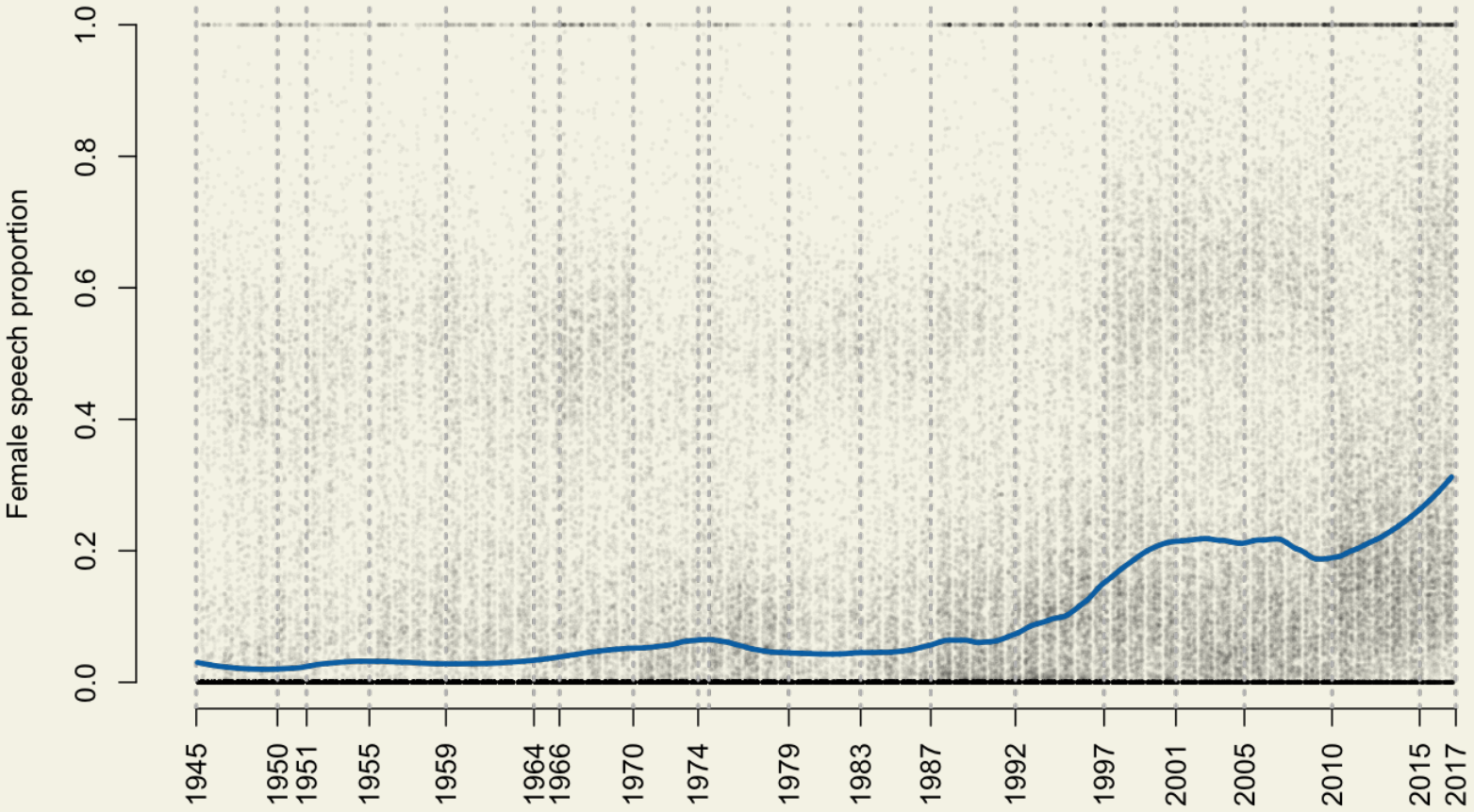
Participation

Influence

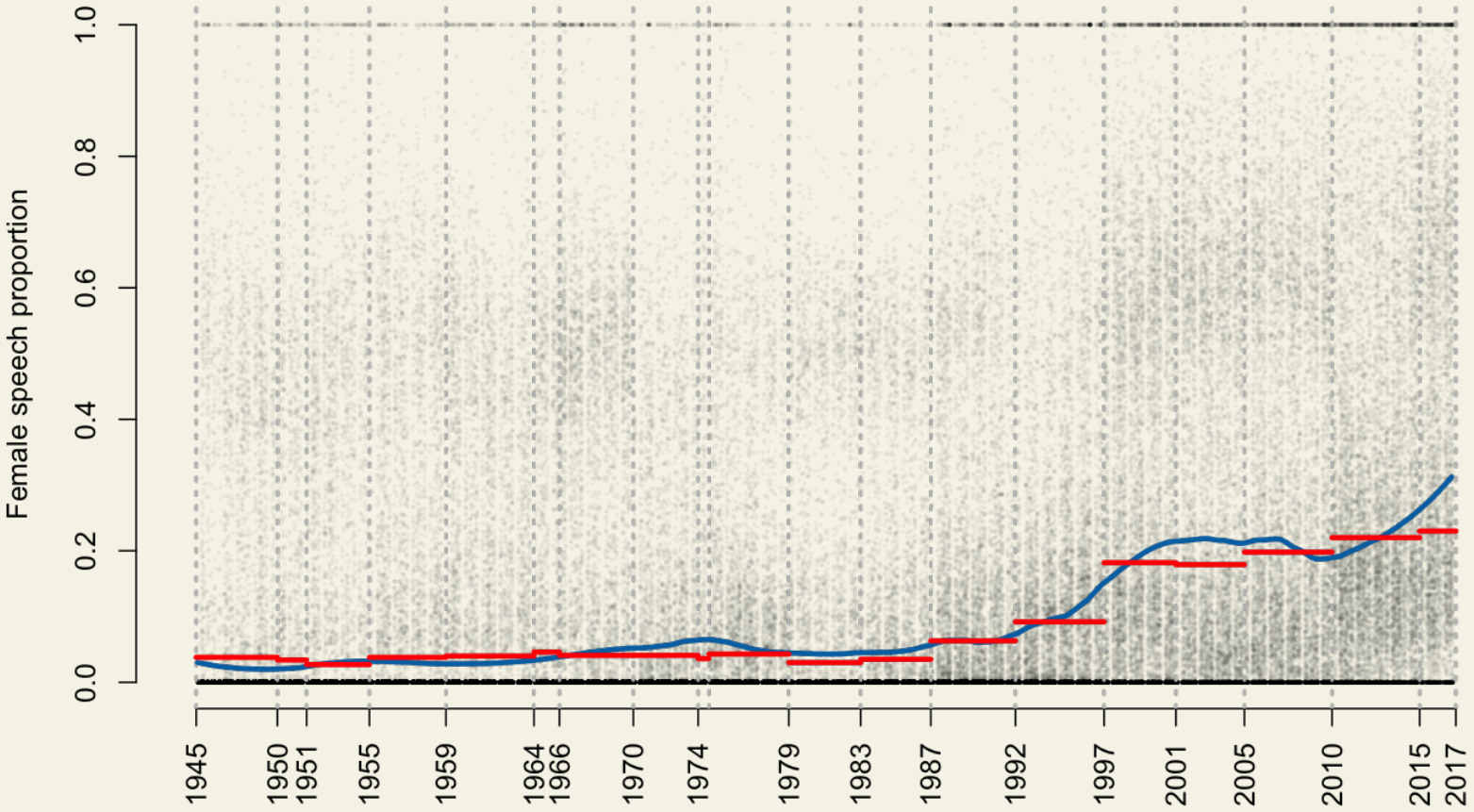
Mechanisms

Discussion

Female speech in the Commons



Female speech in the Commons



Cabinet ministers as role models

Cabinet ministers as role models

- Female role models in electoral politics, education, and business.

Wolbrecht & Campbell, 2007; Beaman et al., 2008, 2012; Gilardi, 2015; Dee, 2007; Nixon & Robinson, 1999; Rask & Bailey, 2002; Bettinger & Long, 2005; Brajer & Gill, 2010; Wang & Kelan, 2013; Bertrand et al., 2014

Cabinet ministers as role models

- Female role models in electoral politics, education, and business.
- Signalling effects

Sapiro, 1981; Mansbridge, 1999; Eagly and Johnson, 1990; Dasgupta and Asgari, 2004; Escobar-Lemmon and Taylor-Robinson, 2008; Krook and O'Brien, 2012; O'Brien, 2015; Mendez & Osborn, 2010; Fox & Lawless, 2004, 2011

Cabinet ministers as role models

- Female role models in electoral politics, education, and business.
- Signalling effects
- Behavioural effects

Kathlene, 1994; Lovenduski and Norris, 1996; Hawkesworth, 2003; Childs, 2004; Karpowitz and Mendelberg, et. al, 2012, 2014; Eagly and Johnson, 1990; Hannah and Murachver, 1999, 2007;

Cabinet ministers as role models

- Female role models in electoral politics, education, and business.
- Signalling effects
- Behavioural effects

Expectations:

1. Female cabinet ministers will increase the **participation** of other female MPs in parliamentary debate.
2. Female cabinet ministers will increase the **influence** of other female MPs in parliamentary debate.

Outline

Motivation

Data and empirical strategy

Participation

Influence

Mechanisms

Discussion

Data

- Full corpus of parliamentary speeches from 1997 to 2017
- Data collected from `theyworkforyou.com`
  - See `www.github.com/jblumenau/twfyR`
- Full sample: $\approx$ 50,000 debates; $\approx$ 1m speeches
- Debates mapped to ministries by cabinet minister speeches
- Final sample: 13,246 debates; $\approx$ 460,000 speeches

Participation

When women are appointed as cabinet ministers, do other women **participate more** in parliamentary debate?

Participation

Dependent variable:

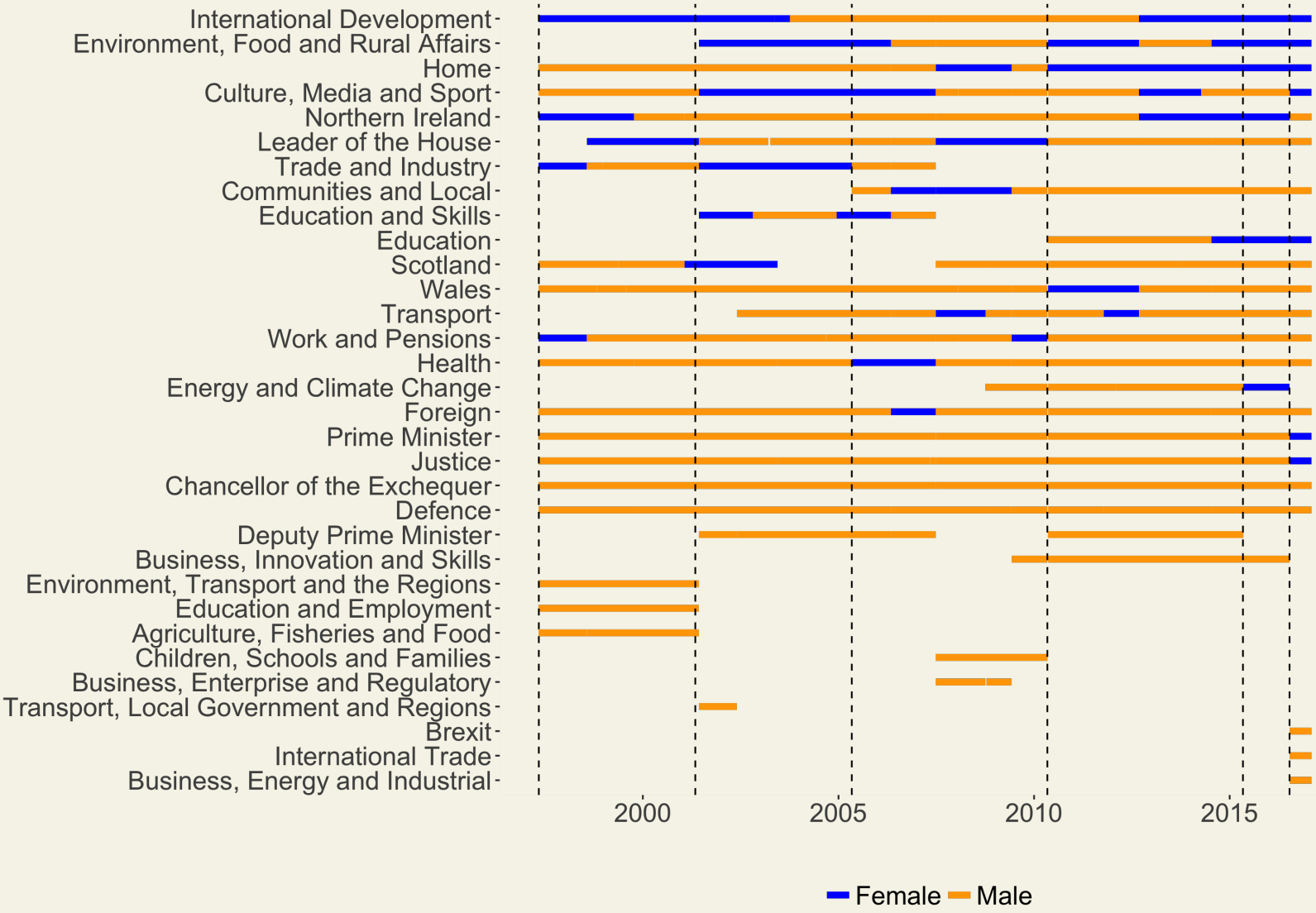
$$PropWordsWomen_{d(mt)} = \frac{\#\text{ words spoken by women}_{d(mt)}}{\#\text{ words spoken by men and women}_{d(mt)}}$$

- m = ministry; t = time period (year-month)
- Speeches by minister responsible for the debate removed

Independent variable:

$$FemaleMinister_{mt} = \begin{cases} 1, & \text{if } Minister_{mt} \text{ is female} \\ 0, & \text{otherwise} \end{cases}$$

$FemaleMinister_{mt}$



Identification

- Female ministers are not randomly allocated across ministries

Identification

- Female ministers are not randomly allocated across ministries
- Female ministers may be more likely to be appointed to ministries where the level of female participation is already high
    - Focus on *within-ministry* variation among ministries that switch between male and female ministers over time

Identification

- Female ministers are not randomly allocated across ministries
- Female ministers may be more likely to be appointed to ministries where the level of female participation is already high
    - Focus on *within-ministry* variation among ministries that switch between male and female ministers over time
- Female ministers are more likely to be appointed when there are more women in parliament
    - Difference out general trends in female participation across ministries in a given month

Identification

- Female ministers are not randomly allocated across ministries
- Female ministers may be more likely to be appointed to ministries where the level of female participation is already high
    - Focus on *within-ministry* variation among ministries that switch between male and female ministers over time
- Female ministers are more likely to be appointed when there are more women in parliament
    - Difference out general trends in female participation across ministries in a given month
- Identification assumption: changes in minister gender are exogenous to female participation, conditional on time and ministry fixed effects

Model specification

$$PropWordsWomen_{d(mt)} = \beta_1 * FemaleMinister_{mt} + \lambda_{m0} + \delta_t + \epsilon_{d(mt)}$$

- $\lambda_{m0} \rightarrow$ ministry fixed effect
- $\delta_t \rightarrow$ time (month-year) fixed effect
- $\beta_1 \rightarrow$ average effect of switching from a male to a female minister amongst ministries that see such a change

Further models to relax the form of the 'parallel trends' assumption:

- Ministry-specific linear, quadratic, and flexible (GAM) time trends details
- These trends purge any smoothly varying, unobserved, ministry specific differences from the estimate of β_1.

Outline

Motivation

Data and empirical strategy

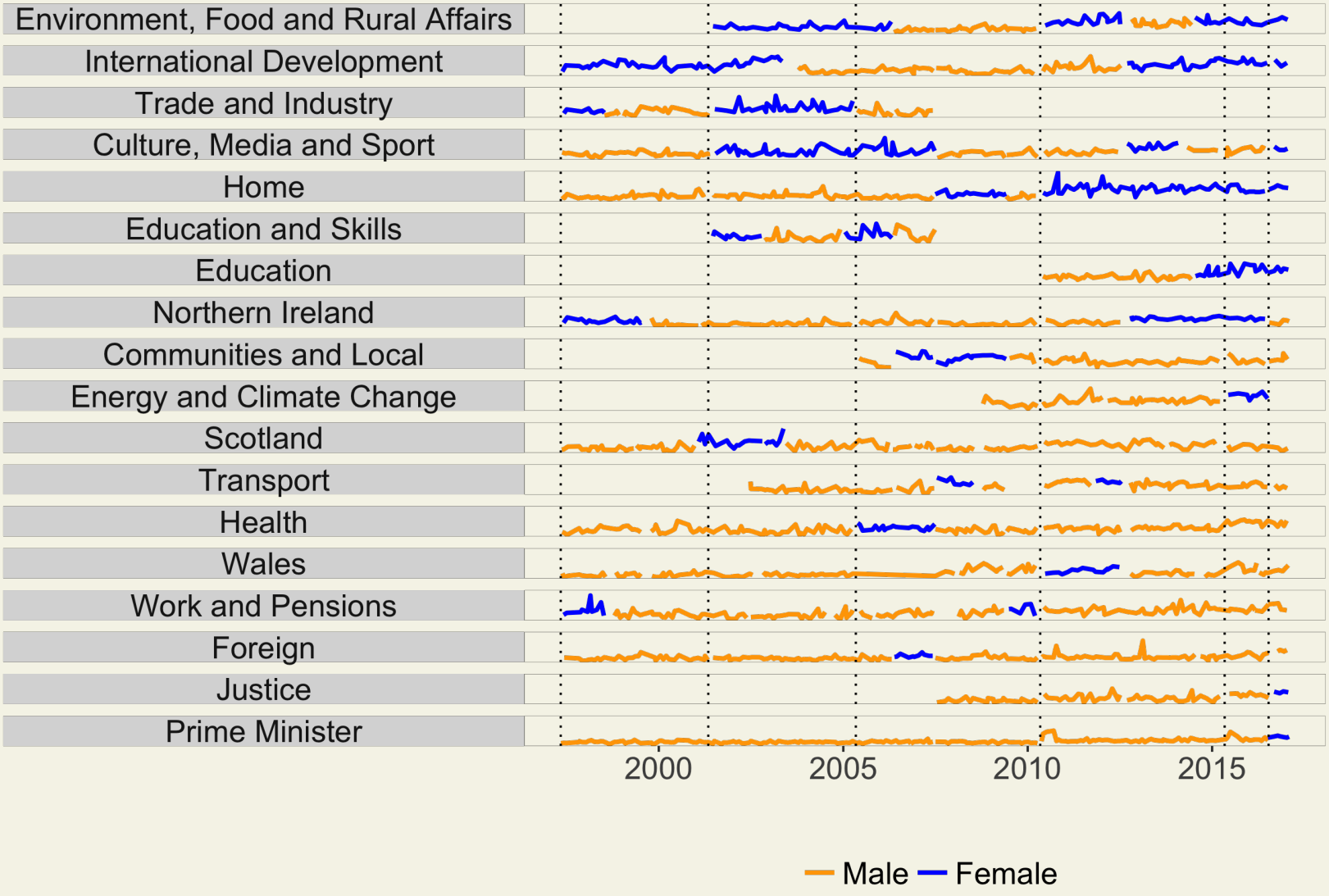
Participation

Influence

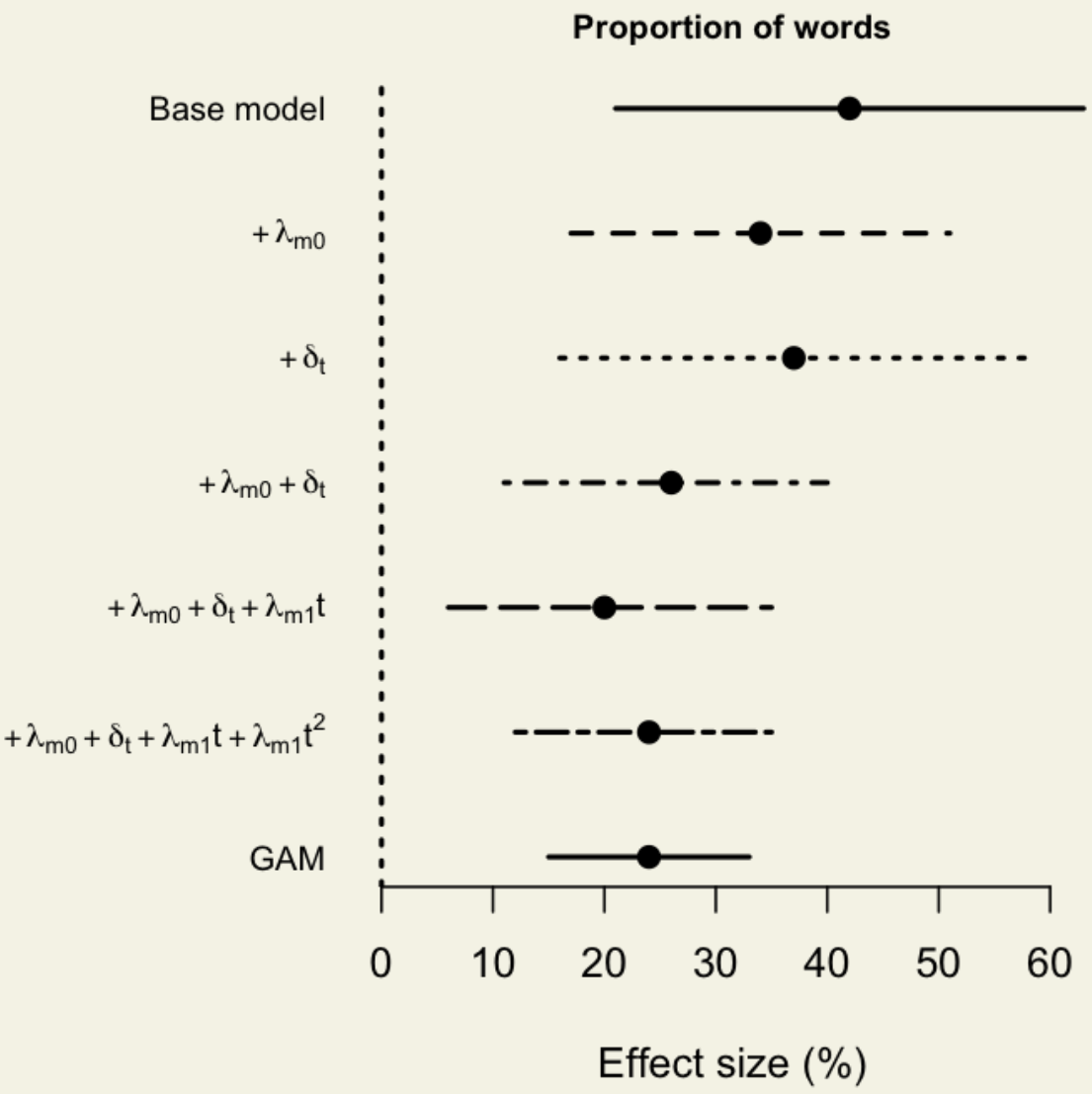
Mechanisms

Discussion

Descriptive analysis

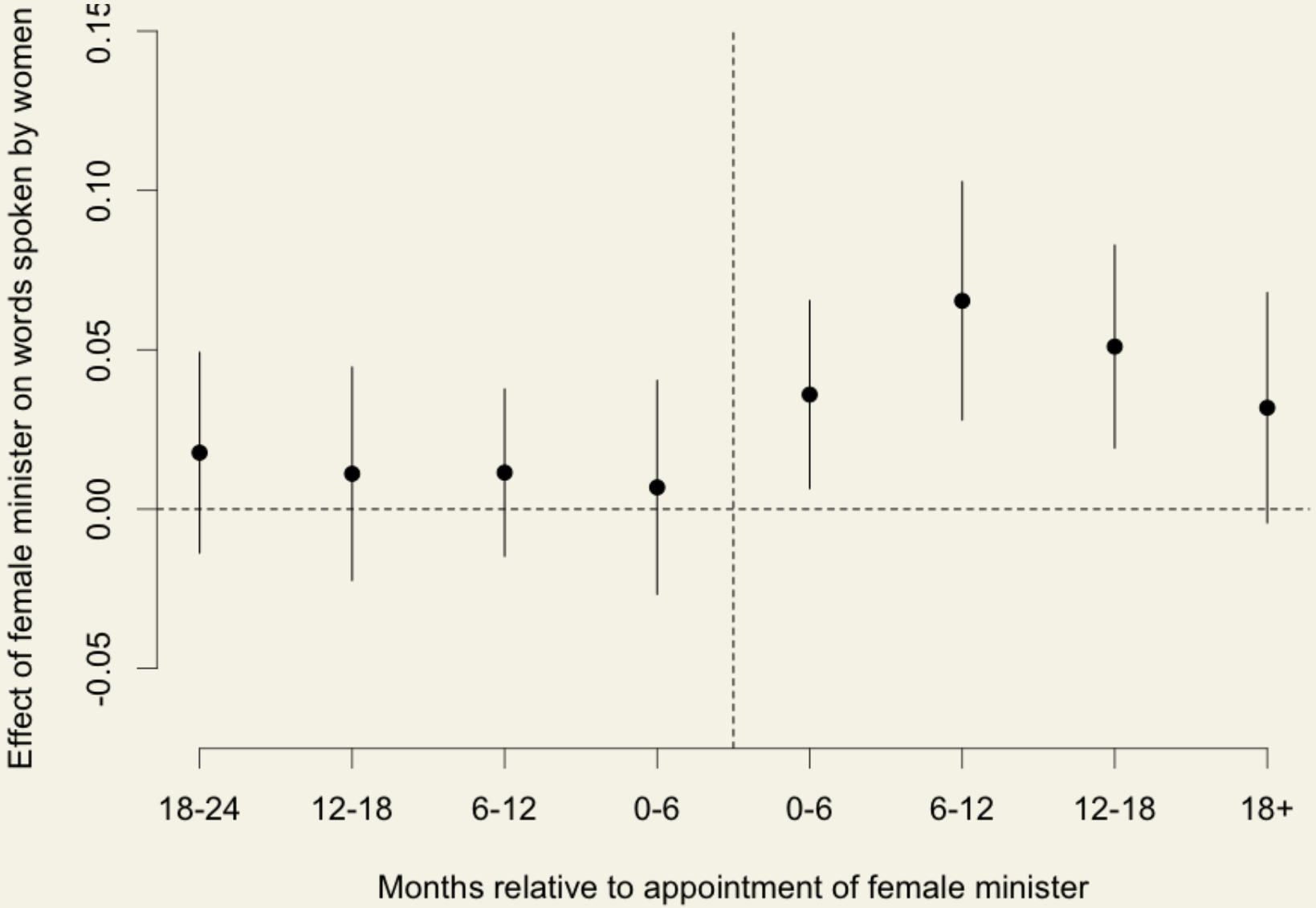


Fixed-effect results



tables

Dynamic panel model



Outline

Motivation

Data and empirical strategy

Participation

Influence

Mechanisms

Discussion

Influence

When women are appointed as cabinet ministers, do other women become **more influential** in parliamentary debate?

Measuring influence

An MP is influential when she **shapes the debate**: when the issues she raises in her speeches are adopted and discussed by other members.

Measuring influence

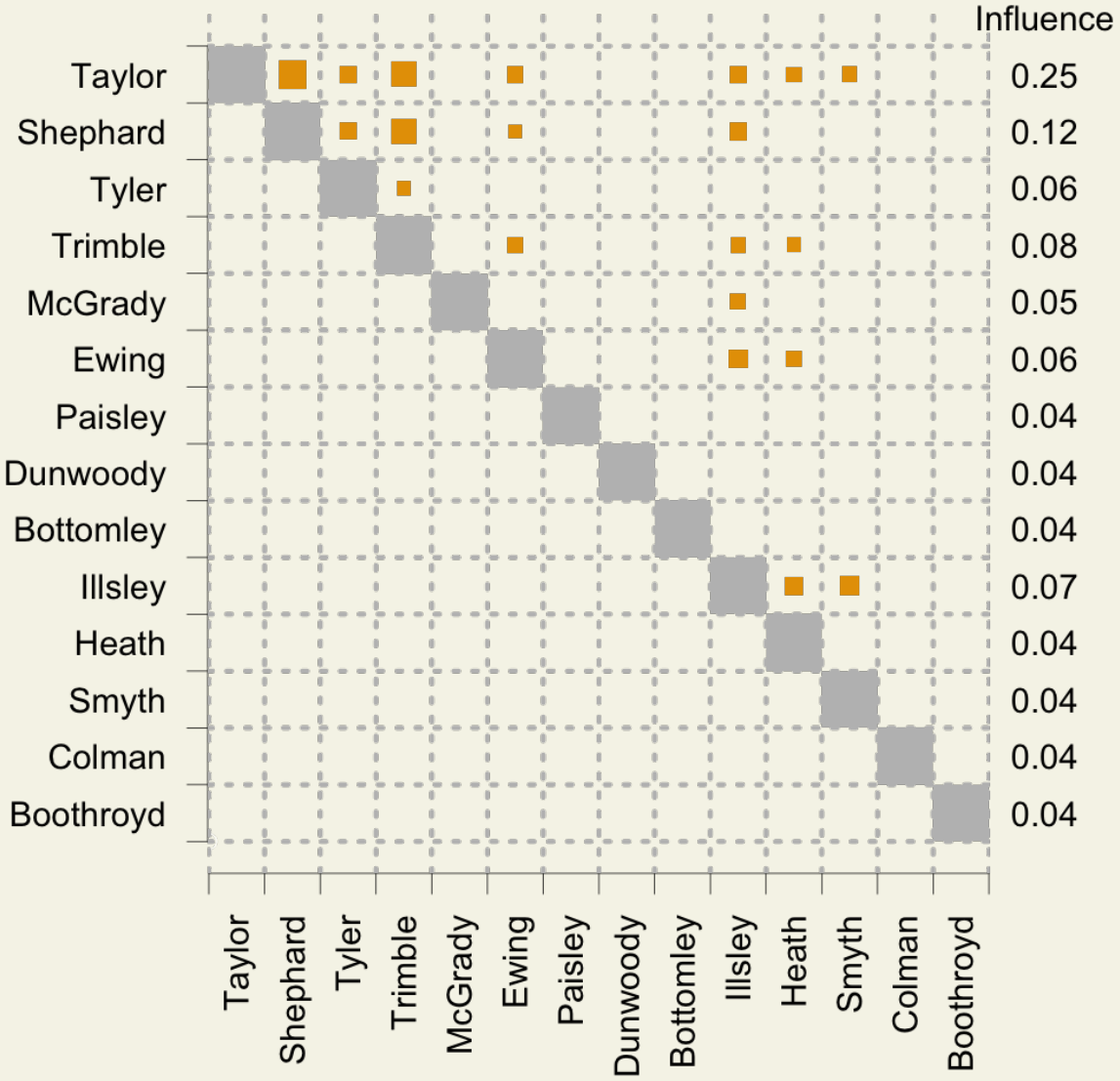
An MP is influential when she **shapes the debate**: when the issues she raises in her speeches are adopted and discussed by other members.

- A speech is influential when it is 'referenced' by many other speeches
- Speech j references speech i if it
  1. occurs after i
  2. uses language that is similar to the language in i
- Influential speeches are referenced by many subsequent speeches
- A speaker delivering more referenced speeches is influential in the debate

Measuring influence

- Each debate is represented as a **directed similarity graph**
- Speeches are **nodes** within the debate graph
  - Speeches are represented as vectors of weighted word counts
  - Speeches are characterised by their most distinctive words
- Two speeches are linked by an **edge** if they use similar words
  - Edges are **weighted** by the cosine-similarity between speech vectors
  - An edge is created only when speech j is after speech i
- A **speech** is more influential the more **central** it is to the network
  - *PageRank* algorithm (Brin and Page, 1999)
- A **speaker** is more influential when she delivers more central speeches

Example


Influence
Taylor 0.25
Shephard 0.12
Tyler 0.06
Trimble 0.08
McGrady 0.05
Ewing 0.06
Paisley 0.04
Dunwoody 0.04
Bottomley 0.04
Illsley 0.07
Heath 0.04
Smyth 0.04
Colman 0.04
Boothroyd 0.04
Taylor
Shephard
Tyler
Trimble
McGrady
Ewing
Paisley
Dunwoody
Bottomley
Illsley
Heath
Smyth
Colman
Boothroyd

Measurement validity

- Ministers are 5 times more influential than backbenchers evidence
- The Speaker of the House is half as influential as backbenchers evidence
- Influence and direct mentions are strongly correlated evidence

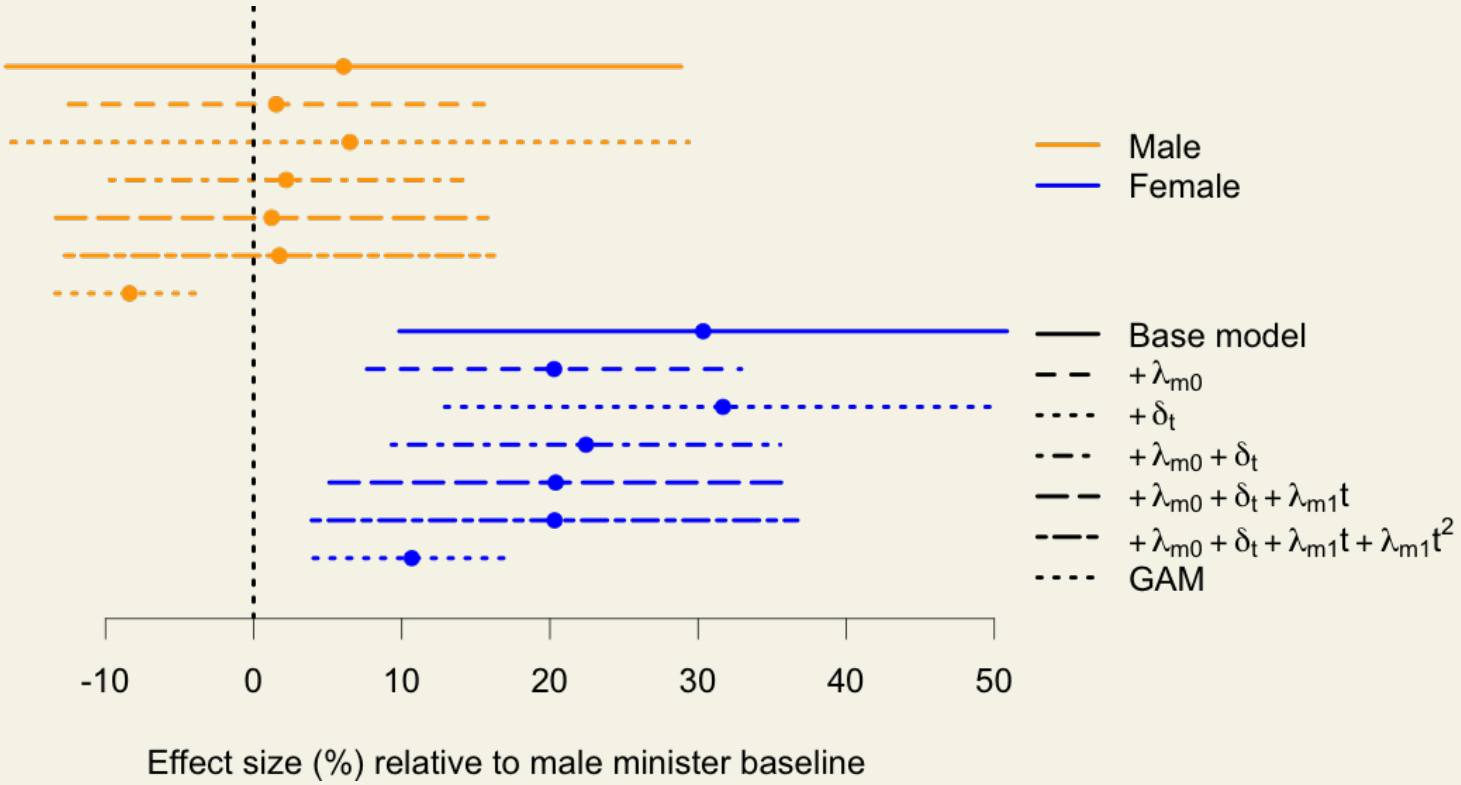
Model specification

$$
\begin{aligned}
influence_{id(mt)} &= \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
&\quad \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
&\quad \lambda_{m0} + \delta_t + \epsilon_{id(mt)}
\end{aligned}
$$

- $influence_{id(mt)}$ is the influence of MP i in debate d (ministry m, time t)
- $\beta_2 \rightarrow$ effect of a female minister on the influence of men
- $\beta_2 + \beta_3 \rightarrow$ effect of a female minister on the influence of women

Expectation: Female ministers will increase female MP influence more than male MP influence ($\beta_2 < \beta_2 + \beta_3$).

Effect of female minister appointment on debate influence



table

Outline

Motivation

Data and empirical strategy

Participation

Influence

Mechanisms

Discussion

Mechanisms

1. Differential agenda-setting by male and female ministers
   - No evidence of an increase in topics traditionally associated with high female participation Details

Mechanisms

1. Differential agenda-setting by male and female ministers
   - No evidence of an increase in topics traditionally associated with high female participation Details

2. Strategic appointment of female shadow ministers
   - No evidence that female cabinet ministers predict female shadow cabinet ministers Details
   - Results robust to excluding shadow cabinet ministers Details

Mechanisms

1. Differential agenda-setting by male and female ministers
   - No evidence of an increase in topics traditionally associated with high female participation Details
2. Strategic appointment of female shadow ministers
   - No evidence that female cabinet ministers predict female shadow cabinet ministers Details
   - Results robust to excluding shadow cabinet ministers Details
3. Differential speechmaking behaviour of female ministers
   - Female ministers are disproportionately responsive to speeches by female MPs Details

Outline

Motivation

Data and empirical strategy

Participation

Influence

Mechanisms

Discussion

Discussion

- When women are promoted to high-office, do they serve as role-models to other women in politics?
  - Cabinet ministers as role-models
- Methodology
  - Fixed-effect design
  - New measures of parliamentary speech
- The appointment of a female cabinet minister increases:
  - the participation of other women by $\approx 25\%$
  - the influence of other women by $\approx 20\%$
- Implications
  - From 'critical mass' to 'critical actors' (Childs and Crook, 2009)
  - Quantifying a broader set of political concepts using text data

Outline

Results tables
- Participation
- Influence
- Responsiveness

Specification details
- Time-trends
- Alternative DV specifications

Additional analyses
- Dynamic panel model
- Individual level results
- Subsets

Validation
- Influence
- Responsiveness

Mechanisms
- Agenda-setting
- Strategic opposition
- Responsiveness

Results – proportion of words

	PropWordsWomen						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female minister	0.073***	0.059***	0.064***	0.044***	0.036***	0.041***	0.042***
	(0.019)	(0.015)	(0.019)	(0.013)	(0.012)	(0.010)	(0.008)
Constant	0.173***	0.093*	0.101	0.035	0.044	0.191	0.087
	(0.011)	(0.048)	(0.087)	(0.054)	(2.765)	(167.177)	(0.196)
Month FEs	×	✓	×	✓	✓	✓	✓
Ministry FEs	×	×	✓	✓	✓	✓	✓
Linear time trends	×	×	×	×	✓	✓	×
Quadratic time trends	×	×	×	×	×	✓	×
Flexible time trends	×	×	×	×	×	×	✓
Effect Size %	42	34	37	26	20	24	24
95% CI	[21,63]	[17,51]	[16,58]	[11,40]	[6,35]	[12,35]	[15,33]
Observations	13,246	13,246	13,246	13,246	13,246	13,246	13,246
R^2	0.016	0.078	0.058	0.109	0.116	0.125	
Adjusted R^2	0.016	0.063	0.056	0.093	0.098	0.105	0.114

back

Results – ratio of words

	RatioWordsWomen						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female minister	0.340***	0.296***	0.284***	0.223***	0.183***	0.209***	0.206***
	(0.078)	(0.076)	(0.067)	(0.061)	(0.064)	(0.053)	(0.041)
Constant	0.853***	0.521*	0.553*	0.199	0.240	1.475	1.027
	(0.049)	(0.271)	(0.328)	(0.296)	(11.261)	(717.353)	(2.847)
Month FEs	×	✓	×	✓	✓	✓	✓
Ministry FEs	×	×	✓	✓	✓	✓	✓
Linear time trends	×	×	×	×	✓	✓	×
Quadratic time trends	×	×	×	×	×	✓	×
Flexible time trends	×	×	×	×	×	×	✓
Effect Size %	40	35	33	26	21	24	24
95% CI	[22,58]	[17,52]	[18,49]	[12,40]	[7,36]	[12,37]	[15,34]
Observations	13,246	13,246	13,246	13,246	13,246	13,246	13,246
R^2	0.014	0.049	0.051	0.082	0.088	0.098	
Adjusted R^2	0.014	0.034	0.049	0.065	0.069	0.077	0.085

back

Results – influence

	Influence						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Constant	0.038***	0.033***	0.048***	0.022	0.041***	−0.855***	35.991
	(0.006)	(0.002)	(0.014)	(0.018)	(0.014)	(0.296)	(164.948)
Female minister	0.003	0.001	0.003	0.001	0.001	0.001	−0.004***
	(0.005)	(0.003)	(0.005)	(0.003)	(0.003)	(0.003)	(0.001)
Female MP	−0.001	−0.0001	0.001	0.001	0.001	0.002	0.001
	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Interaction	0.011***	0.008***	0.011***	0.009***	0.009***	0.008***	0.009***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.001)
Party FEs	✓	✓	✓	✓	✓	✓	✓
Ministry FEs	×	✓	×	✓	✓	✓	✓
Month FEs	×	×	✓	✓	✓	✓	✓
Linear time trends	×	×	×	×	✓	✓	×
Quadratic time trends	×	×	×	×	×	✓	×
Flexible time trends	×	×	×	×	×	×	✓
Observations	174,419	174,419	174,419	174,419	174,419	174,419	174,419
R^2	0.002	0.044	0.016	0.055	0.062	0.065	
Adjusted R^2	0.002	0.044	0.014	0.054	0.060	0.063	0.075

back

Results – responsiveness

	Responsiveness						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	−0.212	0.026	0.683	0.656	0.713	0.757	0.832***
	(1.512)	(1.517)	(1.467)	(1.487)	(1.486)	(1.483)	(0.288)
Female Minister	−0.493	−1.460	−0.581	−1.482***	−1.024**	−0.721	−1.154**
	(0.937)	(1.355)	(0.727)	(0.464)	(0.446)	(0.469)	(0.474)
Interaction	5.738***	5.665***	5.772***	5.660***	5.678***	5.633***	5.390***
	(1.505)	(1.416)	(1.518)	(1.456)	(1.477)	(1.385)	(0.645)
Constant	24.985***	24.629***	31.388***	27.300***	28.794***	186.167*	264.770
	(0.755)	(0.146)	(5.407)	(4.899)	(4.524)	(100.319)	(450.669)
Same Party Control	✓	✓	✓	✓	✓	✓	✓
Ministry FEs	×	✓	×	✓	✓	✓	✓
Month FEs	×	×	✓	✓	✓	✓	✓
Linear time trends	×	×	×	×	✓	✓	×
Quadratic time trends	×	×	×	×	×	✓	×
Flexible time trends	×	×	×	×	×	×	✓
Observations	158,500	158,500	158,500	158,500	158,500	158,500	158,500
R^2	0.004	0.010	0.014	0.018	0.020	0.021	
Adjusted R^2	0.004	0.010	0.013	0.016	0.018	0.019	0.021

back

Outline

Results tables
- Participation
- Influence
- Responsiveness

Specification details
- Time-trends
- Alternative DV specifications

Additional analyses
- Dynamic panel model
- Individual level results
- Subsets

Validation
- Influence
- Responsiveness

Mechanisms
- Agenda-setting
- Strategic opposition
- Responsiveness

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

- t is a time trend variable

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

- t is a time trend variable
- λ_{m1} is a ministry-specific linear time parameter

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

- t is a time trend variable
- λ_{m1} is a ministry-specific linear time parameter
- λ_{m2} is a ministry-specific quadratic time parameter

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

- t is a time trend variable
- λ_{m1} is a ministry-specific linear time parameter
- λ_{m2} is a ministry-specific quadratic time parameter
- f_m denotes flexible (ministry-specific) functions of t

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

- t is a time trend variable
- λ_{m1} is a ministry-specific linear time parameter
- λ_{m2} is a ministry-specific quadratic time parameter
- f_m denotes flexible (ministry-specific) functions of t

Time-trends model specification

$$
\begin{aligned}
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt} \\
Y_{mt} &= \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}
\end{aligned}
$$

- t is a time trend variable
- λ_{m1} is a ministry-specific linear time parameter
- λ_{m2} is a ministry-specific quadratic time parameter
- f_m denotes flexible (ministry-specific) functions of t

Generalized Additive Model (GAM) specification

$$Y_{mt} = \beta_1 * FemaleMinister_{mt} + \lambda_m + \delta_t + \sum_{m=1}^{M} f_m(t) + \epsilon_{mt}$$

where

- f_m denotes arbitrarily-flexible (ministry-specific) functions of t

Alternative specifications for participation

$$PropSpeechesWomen_{mt} = \frac{\#\text{ speeches by women}_{mt}}{\#\text{ speeches by men and women}_{mt}}$$

$$RatioWordsWomen_{mt} = \frac{PropWordsWomen_{mt}}{\text{Proportion of female MPs}_t}$$

$$RatioSpeechesWomen_{mt} = \frac{PropSpeechesWomen_{mt}}{\text{Proportion of female MPs}_t}$$

back

Outline

Results tables
- Participation
- Influence
- Responsiveness

Specification details
- Time-trends
- Alternative DV specifications

Additional analyses
- Dynamic panel model
- Individual level results
- Subsets

Validation
- Influence
- Responsiveness

Mechanisms
- Agenda-setting
- Strategic opposition
- Responsiveness

Dynamic panel model

- We can use the fact that we have data on multiple time periods to check whether "causes happen before consequences, and not vice versa" (Angrist and Pischke, 2009)

Dynamic panel model

- We can use the fact that we have data on multiple time periods to check whether "causes happen before consequences, and not vice versa" (Angrist and Pischke, 2009)
- Conditional on ministry and year fixed-effects, and ministry-specific time trends, is there any statistical difference between treatment and control groups before the treatment happens?

Dynamic panel model

- We can use the fact that we have data on multiple time periods to check whether "causes happen before consequences, and not vice versa" (Angrist and Pischke, 2009)
- Conditional on ministry and year fixed-effects, and ministry-specific time trends, is there any statistical difference between treatment and control groups before the treatment happens?

Dynamic panel model

- We can use the fact that we have data on multiple time periods to check whether "causes happen before consequences, and not vice versa" (Angrist and Pischke, 2009)
- Conditional on ministry and year fixed-effects, and ministry-specific time trends, is there any statistical difference between treatment and control groups before the treatment happens?

$$
\begin{aligned}
Y_{mt} \quad = \quad & \sum_{\tau=0}^{p} \gamma_{-\tau} FemaleMinister_{m,t-\tau} \\
& + \sum_{\tau=1}^{q} \gamma_{+\tau} FemaleMinister_{m,t+\tau} \\
& \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt}
\end{aligned}
$$

- q is the number of lead effects (anticipatory)

back

Dynamic panel model

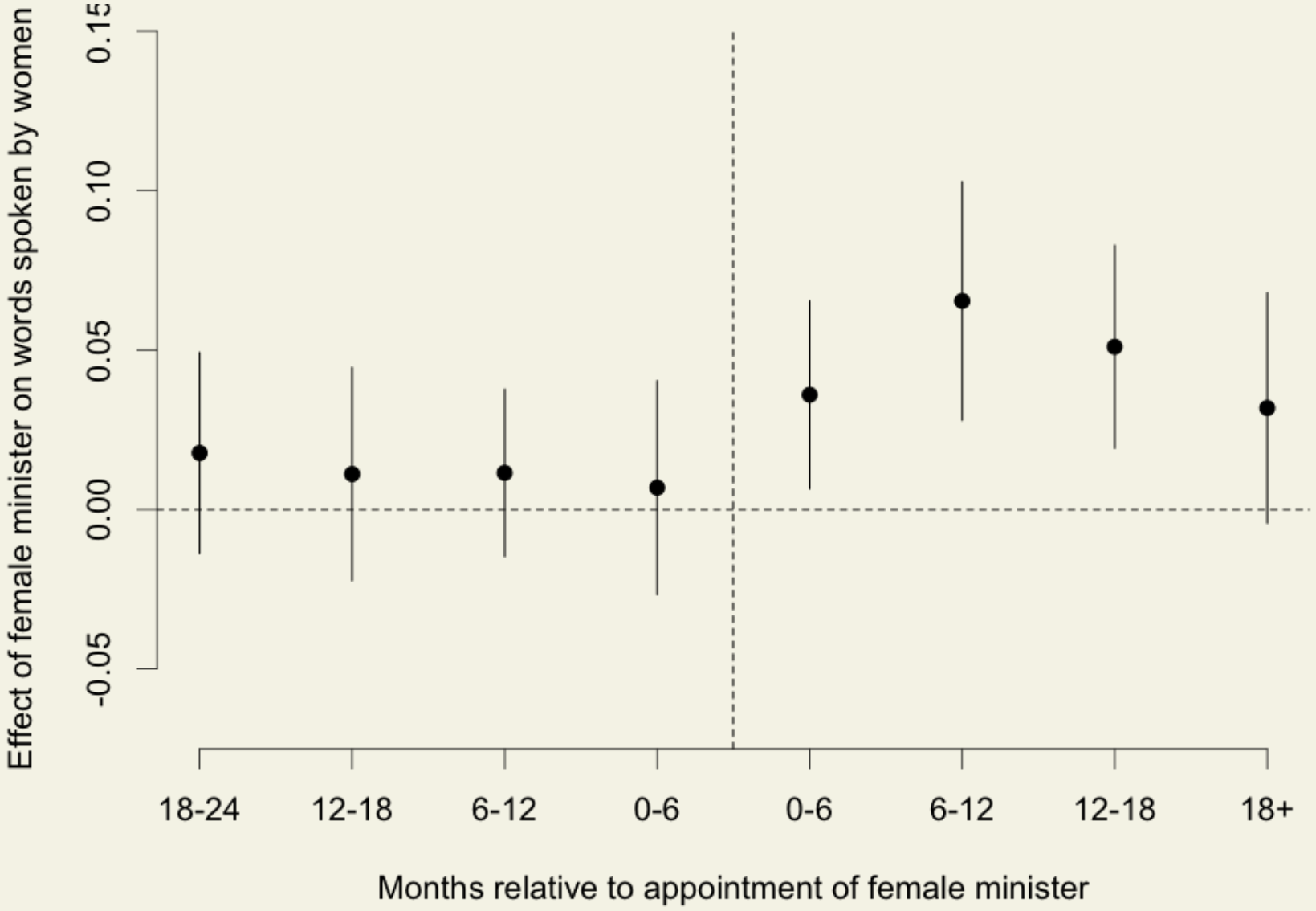
- We can use the fact that we have data on multiple time periods to check whether "causes happen before consequences, and not vice versa" (Angrist and Pischke, 2009)
- Conditional on ministry and year fixed-effects, and ministry-specific time trends, is there any statistical difference between treatment and control groups before the treatment happens?

$$
\begin{aligned}
Y_{mt} \quad = \quad & \sum_{\tau=0}^{p} \gamma_{-\tau} FemaleMinister_{m,t-\tau} \\
& + \sum_{\tau=1}^{q} \gamma_{+\tau} FemaleMinister_{m,t+\tau} \\
& \lambda_m + \delta_t + \sum_{m=1}^{M} \lambda_{m1} t + \sum_{m=1}^{M} \lambda_{m1} t^2 + \epsilon_{mt}
\end{aligned}
$$

- q is the number of lead effects (anticipatory)
- p is the number of lag effects (post-treatment)

back

Dynamic panel model



back

Results – individual level model

back

Participation effect by government

back

Participation effect by party

back

Participation effect by ministry

back

Outline

Results tables
- Participation
- Influence
- Responsiveness

Specification details
- Time-trends
- Alternative DV specifications

Additional analyses
- Dynamic panel model
- Individual level results
- Subsets

Validation
- Influence
- Responsiveness

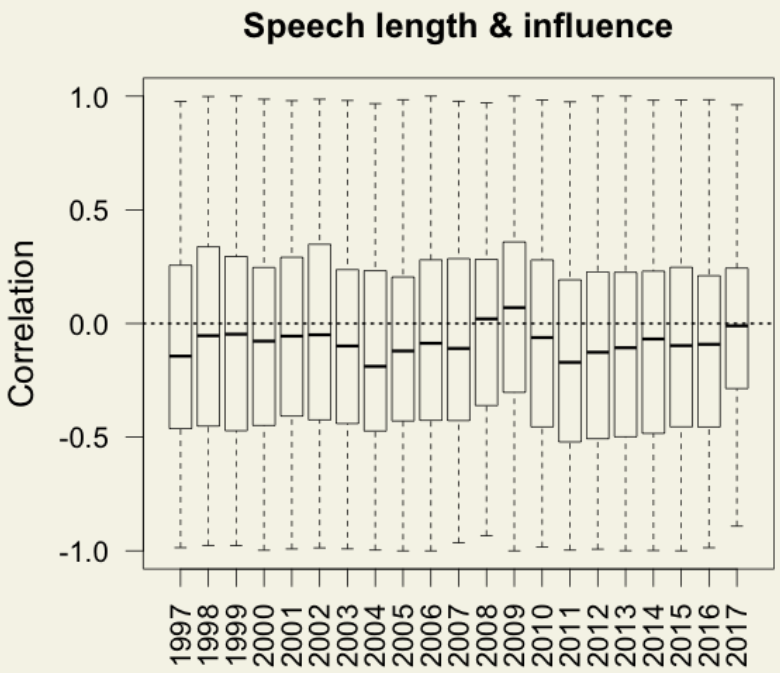
Mechanisms
- Agenda-setting
- Strategic opposition
- Responsiveness

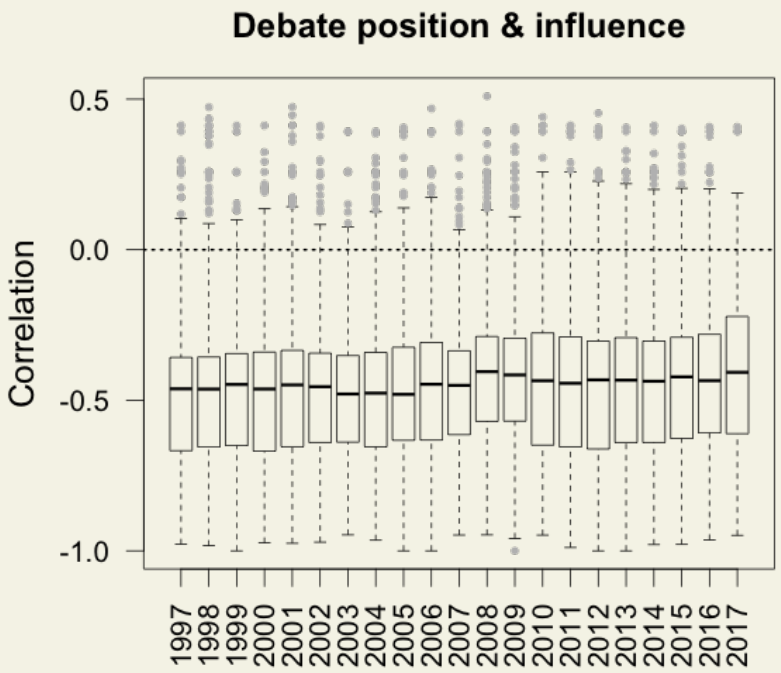
Influence validation

	Influence		
	(1)	(2)	(3)
Minister	0.983***		0.977***
	(0.006)		(0.006)
Speaker		−0.229***	−0.155***
		(0.009)	(0.008)
Constant	0.224***	0.305***	0.230***
	(0.002)	(0.002)	(0.002)
Observations	195,828	195,828	195,828
R^2	0.130	0.004	0.132

back

Influence validation





back

Influence validation

back

Influence validation – direct mentions results

back

Influence validation – controlling for debate position

back

Responsiveness validation – adjacent speeches

	res
Adjacent	6.618***
	(1.197)
Constant	29.629***
	(0.846)
Observations	21,578
R^2	0.001

back

Responsiveness validation – Ministers and backbenchers

	res
Minister responding to backbencher	10.041***
	(0.167)
Constant	9.998***
	(0.121)
Observations	152,577
R^2	0.023

back

Outline

Results tables
- Participation
- Influence
- Responsiveness

Specification details
- Time-trends
- Alternative DV specifications

Additional analyses
- Dynamic panel model
- Individual level results
- Subsets

Validation
- Influence
- Responsiveness

Mechanisms
- Agenda-setting
- Strategic opposition
- Responsiveness

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions

back

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions
    - Correlated Topic Model (Blei and Lafferty, 2006)

back

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions
   - Correlated Topic Model (Blei and Lafferty, 2006)
2. Use these topic proportions as explanatory variables in a series of regressions where the outcome is the level of female participation in a debate

back

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions
   - Correlated Topic Model (Blei and Lafferty, 2006)
2. Use these topic proportions as explanatory variables in a series of regressions where the outcome is the level of female participation in a debate
   - Coefficients indicate the degree to which each latent topic is associated with female participation (under male ministers)

back

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions
   - Correlated Topic Model (Blei and Lafferty, 2006)
2. Use these topic proportions as explanatory variables in a series of regressions where the outcome is the level of female participation in a debate
   - Coefficients indicate the degree to which each latent topic is associated with female participation (under male ministers)
3. Estimate a second set of regressions to discover the relationship between the prevalence of a topic and the gender of the presiding minister

back

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions
   - Correlated Topic Model (Blei and Lafferty, 2006)
2. Use these topic proportions as explanatory variables in a series of regressions where the outcome is the level of female participation in a debate
   - Coefficients indicate the degree to which each latent topic is associated with female participation (under male ministers)
3. Estimate a second set of regressions to discover the relationship between the prevalence of a topic and the gender of the presiding minister
   - Coefficients indicate whether a topic increases or decreases when the minister is female

back

Agenda-setting - measurement

1. Estimate a series of topic models to produce debate-level topic proportions
   - Correlated Topic Model (Blei and Lafferty, 2006)
2. Use these topic proportions as explanatory variables in a series of regressions where the outcome is the level of female participation in a debate
   - Coefficients indicate the degree to which each latent topic is associated with female participation (under male ministers)
3. Estimate a second set of regressions to discover the relationship between the prevalence of a topic and the gender of the presiding minister
   - Coefficients indicate whether a topic increases or decreases when the minister is female
4. Compare the two sets of regression coefficients - if the agenda-setting hypothesis is correct, there will be a positive association between these two coefficients

back

1. Estimate topic models to produce θ: a J x D matrix of topic proportions that describe the fraction of each ministerial statement $d \in \{1, 2, ..., D\}$ that is from each topic $t \in \{1, 2, ..., T\}$.

back

1. Estimate topic models to produce θ: a J x D matrix of topic proportions that describe the fraction of each ministerial statement $d \in \{1, 2, ..., D\}$ that is from each topic $t \in \{1, 2, ..., T\}$.

2. Use θ to predict Y_d (under male ministers).

$$Y_d = b_1\theta_{1d} + b_2\theta_{2d} + ... + b_T\theta_{Td} + \epsilon_d \tag{1}$$

b represents the degree to which each topic is associated with female participation in debates when the minister is male, holding other topics constant.

back

3. Estimate the prevalence of a topic, given the presence of a female minister

$$\theta_{td} = \alpha + \gamma_1 * \mathit{FemaleMinister}_d + \lambda_m + \epsilon_{td} \tag{2}$$

γ represents the change in the use of topic t when a female minister is appointed, conditional on ministry fixed-effects.

back

3. Estimate the prevalence of a topic, given the presence of a female minister

$$\theta_{td} = \alpha + \gamma_1 * \mathit{FemaleMinister}_d + \lambda_m + \epsilon_{td} \tag{2}$$

γ represents the change in the use of topic t when a female minister is appointed, conditional on ministry fixed-effects.

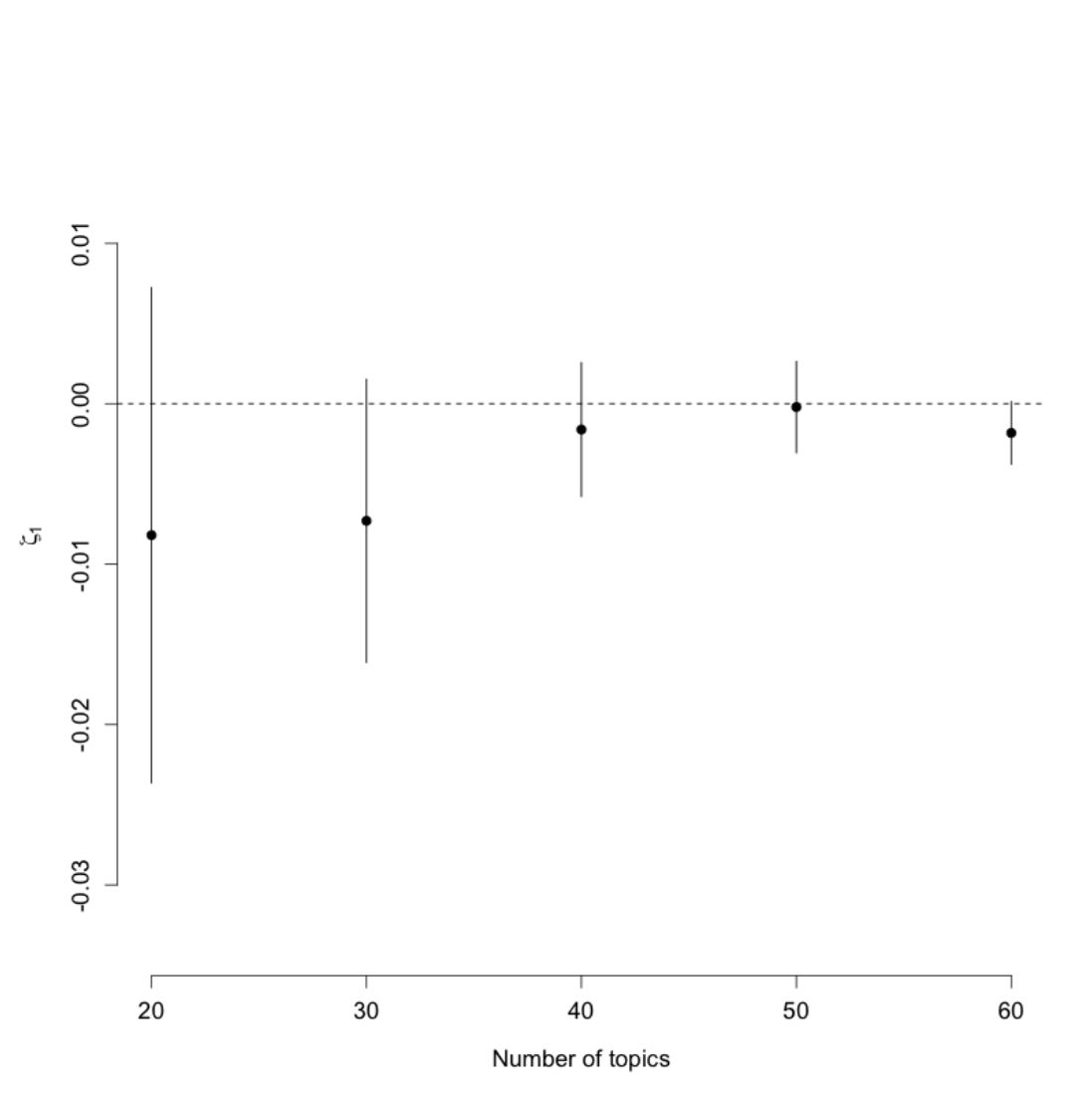
4. Estimate the relationship between b and γ.

$$b_t = \alpha + \zeta_k * \gamma_t + \epsilon_t \tag{3}$$

A positive relationship between these coefficients implies that topics that are associated with high levels of female participation under male ministers become more prevalent when a female minister is appointed.

back

Agenda-setting - results (second stage)



Strategic opposition - results

Probability that a female shadow minister will be appointed, conditional on the sex of the current government minister

	Female shadow minister	
	(1)	(2)
Female government minister	0.141	0.504
	(0.456)	(0.659)
Constant	-1.288^{***}	-0.504
	(0.399)	(1.560)
Ministry FEs	No	Yes
Observations	149	149
Note:	$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	
		Logit model

Details

Strategic opposition - results

Effect of appointing a female minister on female debate participation
(shadow ministers excluded)

	PropWordsWomen						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female minister	0.041***	0.038***	0.037***	0.032***	0.040***	0.038***	0.034***
	(0.013)	(0.013)	(0.010)	(0.011)	(0.011)	(0.011)	(0.008)
Constant	0.173***	0.100**	0.128*	0.058	0.017	3.814	0.339
	(0.009)	(0.046)	(0.077)	(0.073)	(2.022)	(255.854)	(0.425)
Month FEs	×	✓	×	✓	✓	✓	✓
Ministry FEs	×	×	✓	✓	✓	✓	✓
Linear time trends	×	×	×	×	✓	✓	×
Quadratic time trends	×	×	×	×	×	✓	×
Flexible time trends	×	×	×	×	×	×	✓
Effect Size %	24	22	21	19	23	22	20
95% CI	[10,38]	[7,37]	[10,32]	[6,31]	[10,36]	[10,34]	[11,29]
Observations	14,072	14,072	14,072	14,072	14,072	14,072	14,072
R^2	0.005	0.030	0.031	0.053	0.058	0.065	
Adjusted R^2	0.005	0.015	0.029	0.037	0.039	0.044	0.049

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
    1. it occurs directly after i

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer
- Measurement: cosine-similarity between i and j multiplied by the number of words in j: $res_{j \rightarrow i} = sim(v_i, v_j) * n_j$

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer
- Measurement: cosine-similarity between i and j multiplied by the number of words in j: $res_{j \rightarrow i} = sim(v_i, v_j) * n_j$
- Intuition: a speech j is responsive to i when it contains many similar words to i

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer
- Measurement: cosine-similarity between i and j multiplied by the number of words in j: $res_{j \rightarrow i} = sim(v_i, v_j) * n_j$
- Intuition: a speech j is responsive to i when it contains many similar words to i

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer
- Measurement: cosine-similarity between i and j multiplied by the number of words in j: $res_{j \to i} = sim(v_i, v_j) * n_j$
- Intuition: a speech j is responsive to i when it contains many similar words to i

Validation:

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer
- Measurement: cosine-similarity between i and j multiplied by the number of words in j: $res_{j \to i} = sim(v_i, v_j) * n_j$
- Intuition: a speech j is responsive to i when it contains many similar words to i

Validation:

1. Government ministers are ($\approx 200\%$) more responsive to backbenchers than backbenchers are to government ministers table

Responsiveness

"Why do women talk less when talking with men? Conversational partners who offer encouragement and are attentive and responsive are more likely to elicit frequent and active participation from speakers in the conversation." (Hannah & Murachvar, 2007)

- Speech j responds to speech i when:
  1. it occurs directly after i
  2. it engages with the same thematic content as i
  3. it is longer
- Measurement: cosine-similarity between i and j multiplied by the number of words in j: $res_{j \to i} = sim(v_i, v_j) * n_j$
- Intuition: a speech j is responsive to i when it contains many similar words to i

Validation:

1. Government ministers are ($\approx 200\%$) more responsive to backbenchers than backbenchers are to government ministers table
2. Adjacent speeches are ($\approx 40\%$) more responsive than non-adjacent speeches table

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches
- res_{simt} is the response *received* by a speech s made by MP i in ministry m at month t

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches
- res_{simt} is the response *received* by a speech s made by MP i in ministry m at month t
- $\beta_1 \rightarrow$ the difference between responses received by male and female MPs when the minister is male

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches
- res_{simt} is the response *received* by a speech s made by MP i in ministry m at month t
- $\beta_1 \rightarrow$ the difference between responses received by male and female MPs when the minister is male
- $\beta_2 \rightarrow$ the effect of a female minister for the responses received by male MPs

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches
- res_{simt} is the response *received* by a speech s made by MP i in ministry m at month t
- $\beta_1 \rightarrow$ the difference between responses received by male and female MPs when the minister is male
- $\beta_2 \rightarrow$ the effect of a female minister for the responses received by male MPs
- $\beta_2 + \beta_3 \rightarrow$ the effect of a female minister for the responses received by female MPs

Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches
- res_{simt} is the response *received* by a speech s made by MP i in ministry m at month t
- $\beta_1 \rightarrow$ the difference between responses received by male and female MPs when the minister is male
- $\beta_2 \rightarrow$ the effect of a female minister for the responses received by male MPs
- $\beta_2 + \beta_3 \rightarrow$ the effect of a female minister for the responses received by female MPs

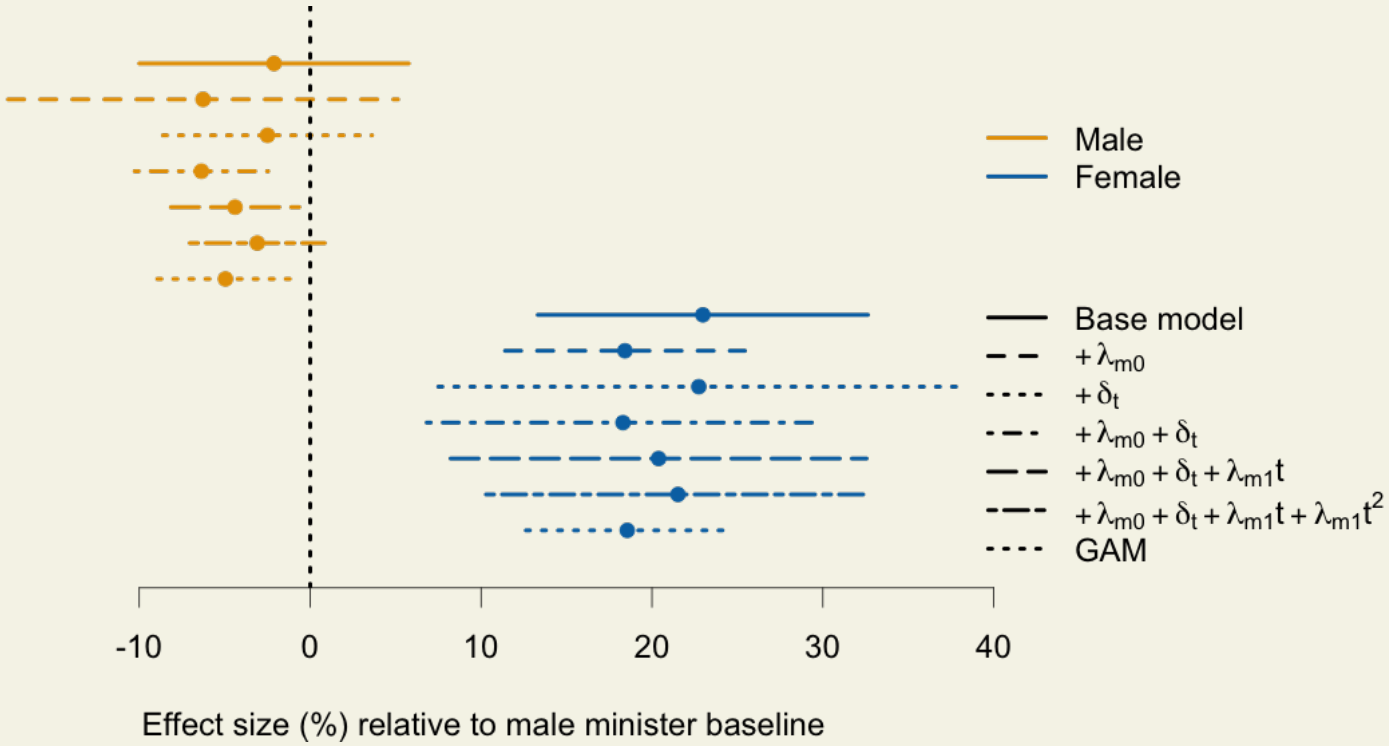
Model specification

$$
\begin{aligned}
res_{simt} \quad = \quad & \beta_1 * FemaleMP_i + \beta_2 * FemaleMinister_{mt} + \\
& \beta_3 * (FemaleMP_i * FemaleMinister_{mt}) + \\
& \lambda_m + \delta_t + \epsilon_{simt} \qquad (4)
\end{aligned}
$$

- Speeches by backbenchers which are immediately followed by ministerial speeches
- res_{simt} is the response *received* by a speech s made by MP i in ministry m at month t
- $\beta_1 \rightarrow$ the difference between responses received by male and female MPs when the minister is male
- $\beta_2 \rightarrow$ the effect of a female minister for the responses received by male MPs
- $\beta_2 + \beta_3 \rightarrow$ the effect of a female minister for the responses received by female MPs

Expectation: $\beta_2 < \beta_2 + \beta_3$.

Effect of female minister appointment on responsiveness



tables